# ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** second | Second. |
| 2 | **(c)** distance / time | Speed formula. |
| 3 | **(c)** uniform | Uniform. |
| 4 | **(c)** 60 seconds | 60 s. |
| 5 | **(c)** m/s | Metre per second. |
| 6 | **(b)** non-uniform | Non-uniform. |
| 7 | **(c)** 60 minutes | 60 min. |
| 8 | **(a)** 50 km/h | 100/2. |
| 9 | **(b)** non-uniform motion | Stops and starts. |
| 10 | **(b)** 24 hours | 24 h. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Distance per unit time (1); Unit is m/s (1). | 1+1 |
| 12 | Time for one oscillation (1); In seconds (1). | 1+1 |
| 13 | Unequal distance in equal time (1); Changing speed (1). | 1+1 |
| 14 | Second (1); Symbol s (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Second (1); Minute (1); Hour (1). | 1+1+1 |
| 16 | Distance = speed x time (1); 40 x 2 (1); 80 km (1). | 1+1+1 |
| 17 | Second hand (1); Fan blades (1); Escalator (1). | 1+1+1 |
| | **OR:** Sundial (1); Water clock (1); Quartz clock (1). | 1+1+1 |
| 18 | Time = distance / speed (1); 150/50 (1); 3 h (1). | 1+1+1 |
| | **OR:** On y-axis (1); With units (1); Against time (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Speed = distance / time (1); 120/20 (1); 6 (1); m/s (1); 6 m/s (1). | 1+1+1+1+1 |
| | **OR:** 18 x 5/18 (1); 5 m/s (1); 10 x 18/5 (1); 36 km/h (1); Use the factor (1). | 1+1+1+1+1 |
| 20 | Sundial (1); Water clock (1); Sand timer (1); Pendulum clock (1); Quartz watch (1). | 1+1+1+1+1 |
| | **OR:** 360/6 (1); 60 km/h (1); 60 x 2 (1); 120 km (1); Uniform speed (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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# ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** 5 m/s | 10/2. |
| 2 | **(b)** uniform speed | Uniform. |
| 3 | **(d)** oscillation | Oscillation. |
| 4 | **(c)** 60 km/h | 60/1. |
| 5 | **(a)** rest | No change in distance. |
| 6 | **(a)** time period | Period. |
| 7 | **(b)** 60 km/h | 240/4. |
| 8 | **(d)** x-axis (horizontal) | Horizontal. |
| 9 | **(d)** pendulum or quartz | Time keeping. |
| 10 | **(b)** 5 m/s | 200/40. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | 80/2 (1); 40 km/h (1). | 1+1 |
| 12 | Clock (1); Stopwatch (1). | 1+1 |
| 13 | A bob hung from a string (1); Swings to and fro (1). | 1+1 |
| 14 | 20/4 (1); 5 m/s (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Pendulum (1); Clock hands (1); Swing (1). | 1+1+1 |
| 16 | 2 x 60 = 120 min (1); 120 x 60 (1); 7200 s (1). | 1+1+1 |
| 17 | For meaning (1); For comparison (1); For clarity (1). | 1+1+1 |
|  | **OR:** Bus in traffic (1); Falling leaf (1); Running dog (1). | 1+1+1 |
| 18 | 3 x 60 (1); 180 (1); 180 s (1). | 1+1+1 |
|  | **OR:** Count 10 oscillations (1); Note the time (1); Divide by 10 (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Hang a bob (1); Pull and release (1); Count 20 swings (1); Note time (1); Divide by 20 (1). | 1+1+1+1+1 |
|  | **OR:** Reset (1); Start on time (1); Stop on time (1); Avoid delay (1); Repeat (1). | 1+1+1+1+1 |
| 20 | Fan: uniform (1); Clock hand: uniform (1); Bus in traffic: non-uniform (1); Falling leaf: non-uniform (1); Kite: non-uniform (1). | 1+1+1+1+1 |
|  | **OR:** Pendulum (1); Swing (1); Clock hand (1); Earth's rotation (1); Moon's orbit (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** y-axis (vertical) | Vertical. |
| 2 | **(c)** shadow | Shadow. |
| 3 | **(a)** 60 km | 30 x 2. |
| 4 | **(c)** swinging swing | Repeats. |
| 5 | **(c)** speedometer | Speed. |
| 6 | **(b)** 150 km | 50 x 3. |
| 7 | **(a)** falling stone | Does not repeat. |
| 8 | **(a)** odometer | Distance. |
| 9 | **(c)** 50 m | 10 x 5. |
| 10 | **(b)** stopwatch | Time. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Distance travelled (1); In a vehicle (1). | 1+1 |
| 12 | One to and fro swing (1); Of a pendulum (1). | 1+1 |
| 13 | Equal distance in equal time (1); Constant speed (1). | 1+1 |
| 14 | Speed (1); In a vehicle (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Speed = distance / time (1); 90/3 (1); 30 km/h (1). | 1+1+1 |
| 16 | Races (1); Experiments (1); Cooking (1). | 1+1+1 |
| 17 | Speed = distance / time (1); Distance = speed x time (1); Time = distance / speed (1). | 1+1+1 |
| | **OR:** 300/60 (1); 5 (1); 5 m/s (1). | 1+1+1 |
| 18 | 20/10 (1); 2 (1); 2 s (1). | 1+1+1 |
| | **OR:** m/s (1); km/h (1); cm/s (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Total 120 km (1); Time 2 h (1); Average 120/2 (1); 60 (1); 60 km/h (1). | 1+1+1+1+1 |
| | **OR:** Travel time (1); Safety (1); Racing (1); Fuel use (1); Planning (1). | 1+1+1+1+1 |
| 20 | Axes labelled (1); Points (1,20), (2,40) (1); (3,60), (4,80) (1); Join points (1); Straight line (1). | 1+1+1+1+1 |
| | **OR:** 1 min = 60 s (1); 1 h = 60 min (1); 1 day = 24 h (1); 1 year = 365 days (1); SI unit is second (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.